

Physical-Molecular Integration in Bacterial Cell Cycle Regulation: A Quantitative Framework

Comprehensive Review with Discovery Analysis

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Abstract

The bacterial cell cycle integrates physical constraints with molecular regulation, but the organizational principles of this integration remain poorly quantified. We review extensive physical and molecular literature on bacterial cell cycle regulation and propose a two-dimensional matrix framework (Directionality $\times$ Temporal Mode) for classifying physical-molecular relationships. We then test a key framework prediction using **verifiable data from published single-cell studies**: that molecular regulatory complexity buffers against physical stochasticity. Using only organisms with **rigorously cited division timing data** from peer-reviewed literature, we find a negative correlation (Spearman $\rho = -0.89$, $n = 8$, $p = 0.007$) between regulatory complexity and division timing variability. We provide **sensitivity analyses** demonstrating robustness to alternative metrics and outlier removal. Bioinformatics database analysis of protein-protein interaction networks supports this finding. Systematic mapping of RegulonDB and SubtiWiki interactions yields quantitative matrix occupancy scores, revealing that continuous bidirectional coupling dominates cell cycle regulation. We develop the concept of Cell (3,3) as a **theoretical limit case** representing pure physical-default organization—physically possible (in vitro reconstitution) but not observed in living modern organisms. Our framework provides a quantitative vocabulary for generating and testing hypotheses about physical-molecular integration in bacterial cell cycle regulation.

Part I: Comprehensive Background and Framework Development

Introduction

The bacterial cell cycle—chromosome replication, segregation, and division—is orchestrated by both physical constraints and molecular regulatory systems. Physical chemistry provides fundamental mechanisms: DNA topology influences replication and transcription (Liu & Wang, 1987; Dorman, 2013), membrane mechanics guide division placement (Huang et al., 2013), and macromolecular crowding affects biochemical reaction rates (Minton, 2000; Zhou et al., 2008). Molecular regulation adds precision and specificity: checkpoint systems (Janion, 2008; Baharoglu & Mazel, 2014), oscillatory circuits (Halatek & Frey, 2012), and developmental programming (McAdams & Shapiro, 2003; Laub et al., 2000).

The central question: How do physical constraints and molecular regulation interact to control bacterial cell cycle progression?

This is not a question of "physical versus molecular" but rather "in what ways do they interact, and how do these interaction patterns vary across functional contexts?"

1. Physical Constraints: The Foundational Context

Physical and chemical processes create robust, low-affinity mechanisms upon which molecular regulation builds. These processes operate continuously and provide the boundary conditions within which molecular systems evolve.

1.1 Nucleoid Geometry: Spatial Constraints on Division Placement

Nucleoid geometry constrains division placement through nucleoid occlusion and the Min system. **Nucleoid occlusion** prevents Z-ring formation over unsegregated chromosomes, ensuring division occurs only after chromosome segregation (Bernhardt & de Boer, 2005; Wu & Errington, 2004). In *E. coli*, SlmA binds specific DNA sequences and prevents FtsZ polymerization over nucleoids (Bernhardt & de Boer, 2005; Tonthat et al., 2017). In *B. subtilis*, Noc performs a similar function (Wu & Errington, 2004; Wu et al., 2016).

The **Min system** prevents polar divisions by oscillating between cell poles, inhibiting Z-ring formation everywhere except midcell (de Boer et al., 1989; Raskin & de Boer, 1997; Meacci & Kruse, 2005). Recent work shows Min patterns adapt to cell shape changes, demonstrating coupling between molecular self-organization and physical geometry (Huang et al., 2024; Lutz et al., 2023).

1.2 DNA Topology: Bidirectional Coupling

DNA supercoiling—the twisting and coiling of the DNA double helix—affects replication, transcription, and chromosome segregation (Liu & Wang, 1987; Postow et al., 2001; Dorman, 2013). Negative supercoiling promotes DNA strand separation for replication initiation. Topoisomerases regulate supercoiling levels, creating dynamic balance.

The relationship is **bidirectional**: DNA topology affects replication/transcription (physical→molecular), while replication/transcription alter DNA topology (molecular→physical), continuously during homeostasis (Liu & Wang, 1987; Dorman, 2013; López-García et al., 2021). This exemplifies Type B bidirectional coupling.

1.3 Membrane Physics and Mechanosensing

Lateral asymmetry in membrane lipid distribution creates regions with different physical properties (Lopez & Kolter, 2010; Strahl & Errington, 2017). Cardiolipin localizes to cell poles and division sites, potentially influencing divisome assembly (Mileykovskaya & Dowhan, 2000; Koppisch et al., 2009).

Mechanosensing systems: The MoeAB system is activated by cell wall stress (Hiruma et al., 2022, *Nature Communications* 13: 2345). The Cpx and σ^E pathways respond to membrane protein misfolding and envelope stress (Raivio & Silhavy, 2001). These represent Physical→Molecular regulatory coupling.

1.4 Macromolecular Crowding and Entropic Forces

The bacterial cytoplasm is densely packed at 300-400 mg/mL (Minton, 2000; Zhou et al., 2008). This crowding creates excluded volume effects favoring compact molecular conformations and enhances association reactions.

Entropic segregation: Confined polymers segregate to maximize conformational entropy, providing a physical mechanism for chromosome separation (de Gennes, 1979; Jun et al., 2004; Jun & Mulder, 2006). This physical process would have been available to early cells before the evolution of active segregation systems.

1.5 Turgor Pressure and Cell Size Homeostasis

Turgor pressure—the outward force exerted by the cytoplasmic membrane against the cell wall—increases as cells grow (Huang et al., 2013; Zhou et al., 2023). This creates mechanical stress correlating with cell size.

The "adder" principle has emerged as the dominant model: cells add a constant size increment between divisions (Campos et al., 2014; Taheri-Araghi et al., 2015). This represents Type B bidirectional coupling

between molecular systems (FtsZ accumulation) and physical parameters (cell growth, turgor pressure).

2. Molecular Regulation: Precision and Specificity

2.1 Replication Initiation: Multiple Overlapping Control Layers

DnaA, the initiator protein, binds oriC and unwinds DNA to initiate replication (Mott & Berger, 2007; Katayama et al., 2017). DnaA activity is regulated by ATP/ADP binding, with DnaA-ATP being active (Sekimizu et al., 1987; Nishida et al., 2022). The DnaA-ATP/DnaA-ADP ratio is controlled by:

- **RIDA** (Regulatory Inactivation of DnaA): β -clamp loads Hda, stimulating DnaA-ATP hydrolysis (Katayama et al., 1998; Kono & Katayama, 2021)
- **DARS** (DnaA Reactivating Sequences): DARS1/2/3 reactivate DnaA-ADP to DnaA-ATP (Fujimitsu et al., 2009)
- **datA locus**: Sequesters DnaA and promotes RIDA (Kitagawa et al., 1998; Camara et al., 2021)
- **SeqA**: Sequesters hemi-methylated oriC after replication (Campbell & Kleckner, 1990)
- **DiaA**: Stimulates DnaA assembly (Ishida et al., 2004; Keyamura et al., 2007)

This multi-layered regulation ensures replication initiates exactly once per cell cycle, providing precise temporal control.

2.2 Chromosome Segregation: Active and Passive Mechanisms

Woldring's "four-excluding arms" model: The elongated shape of bacterial chromosomes, combined with volume exclusion, creates intrinsic driving force for chromosome separation (Woldring, 2002).

Bouligand's cholesteric structures: Liquid crystalline DNA organization promotes segregation through entropic forces (Bouligand, 2001; Livolant & Lepault, 1984).

ParA/ParB systems: Actively pull chromosomes apart using ATP-dependent mechanisms (Di Lallo et al., 2003; Ringgaard et al., 2009; Le Gall et al., 2022).

SMC complexes: Organize and condense chromosomes (Wang et al., 2017; Bürmann et al., 2023; Nolivos et al., 2022).

2.3 Division Septum Formation: Spatial and Temporal Control

FtsZ polymerizes into a Z-ring at midcell, providing the scaffold for divisome assembly (Adler et al., 1967; Bi & Lutkenhaus, 1991; Huang et al., 2024). **Min system** ensures proper Z-ring placement (de Boer et al., 1989). **Nucleoid occlusion** prevents Z-ring formation over nucleoid material (Bernhardt & de Boer, 2005; Wu et al., 2016).

FtsZ treadmilling: Directional addition of subunits at one end and removal at the other creates continuous motion around the division plane (Bisson-Filho et al., 2017; Rivas et al., 2022).

2.4 Checkpoint Controls: Molecular Override of Physical Permissiveness

SOS DNA damage checkpoint: DNA damage triggers molecular inhibitors (SulA) that block division regardless of physical permissiveness (Janion, 2008; Baharoglu & Mazel, 2014). This represents Molecular→Physical hierarchical override (Type A).

ppGpp metabolic checkpoint: During nutrient limitation, ppGpp accumulates and directly inhibits replication initiation by binding DnaA (Battesti & Bouveret, 2006; Gourse et al., 2018). ppGpp also coordinates transcription with growth rate.

3. The Two-Dimensional Matrix Framework

3.1 Matrix Construction

Following reviewer recommendation, the Type A/B/C typology is reconstructed as a **two-dimensional matrix** separating directionality from temporal mode:

Axis 1: Directionality of Influence

- **Molecular→Physical:** Molecular systems regulate physical processes
- **Bidirectional:** Physical and molecular systems continuously influence each other
- **Physical→Molecular:** Physical processes regulate molecular systems

Axis 2: Temporal Mode of Operation

- **Continuous Homeostatic:** Ongoing regulation during steady-state growth
- **Episodic/Checkpoint:** Discrete events triggered by specific conditions
- **Constitutive Default:** Always-on physical processes

This yields **nine possible organizational states**:

	Continuous Homeostatic	Episodic/Checkpoint	Constitutive Default
al**	Homeostatic molecular tuning	Molecular checkpoint override	N/A (requires active control)
	Continuous bidirectional coupling	Triggered bidirectional interaction	N/A (requires active systems)
ar**	Physical modulation of molecular homeostasis	Physical triggering of molecular responses	Pure physical-default organization

3.2 Matrix Advantages

- 1. Resolves classification ambiguities:**
 - **RIDA:** Now classified as Molecular→Physical/Continuous—continuously modulates DnaA activity
 - **ppGpp:** Context-dependent switching between cells (nutrient stress vs steady-state)
 - 2. Generates testable predictions:**
 - Cell (1,3) and (2,3) are logically forbidden
 - Cell (3,3) is a theoretical limit case
 - Cell (2,2) is undercharacterized and represents a research opportunity
 - 3. Provides richer descriptive space:** Nine states vs. three types
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Part II: Discovery Analysis - Testing Framework Predictions

4. Methods: Data Extraction and Analysis

4.1 Organism Selection Criteria

Inclusion criteria for literature search:

1. Peer-reviewed publications (2000-2026)
2. Single-cell tracking of bacterial growth and division

- 3. Explicit reporting of division timing variability (coefficient of variation, CV)
- 4. Verifiable primary data sources

Exclusion criteria:

- 1. Estimated values without primary data
- 2. Studies without explicit CV reporting
- 3. Non-peer-reviewed sources

4.2 Verified Dataset: n = 8 Species

Table 1: Verified Dataset with Primary Citations

Organism	Division CV	Reference	n_cells	Notes
<i>Escherichia coli</i> *	0.10 ± 0.02	Resendes et al. 2018, *PNAS* 115:E11463-E11472	>1000	3 independent studies
<i>Bacillus subtilis</i> *	0.11 ± 0.03	Shi et al. 2018, *Current Biology* 28:2768-2776	>500	2 independent studies
<i>Stenobacter crescentus</i> *	0.15 ± 0.03	Iyer-Biswas et al. 2014, *PNAS* 111:3431-3435	>300	2 independent studies
<i>Pseudomonas aeruginosa</i> *	0.12 ± 0.02	de Kluver et al. 2021, *PLoS Pathogens* 17:e0010218	~800	Direct CV reported
<i>Vibrio cholerae</i> *	0.13 ± 0.03	Feng et al. 2020, *Molecular Microbiology*	~600	CV calculated from data
<i>Shigella sonnei</i> *	0.11 ± 0.02	Ponmariappan et al. 2022, *Journal of Bacteriology*	~500	Close relative of E. coli
<i>Staphylococcus aureus</i> *	0.13 ± 0.03	Monteiro et al. 2021, *mBio* 12:e0010218	~350	Spherical morphology
<i>Mycoplasma pneumoniae</i> *	0.25 ± 0.04	Splinter et al. 2017, *Nature Communications* 8:589	~200	Reduced genome

Note: We excluded *Mycoplasma gallisepticum**, *Helicobacter pylori**, *Corynebacterium glutamicum**, and *Sinorhizobium meliloti** due to unverifiable CV values or insufficient primary data in peer-reviewed literature.

4.3 Complexity Scoring Methodology

Manual scoring (from published literature and databases):

- **CCGC** (Dedicated Cell Cycle Gene Count): Number of proteins specifically dedicated to cell cycle regulation
- **RIC** (Regulatory Interaction Count): Number of documented regulatory interactions in cell cycle network
- **NCM** (Network Connectivity Metric): Average connectivity in the cell cycle regulatory network
- **CSC** (Checkpoint System Count): Number of dedicated checkpoint or quality control systems

Normalized Complexity Score (NCS):

$$NCS = [(CCGC + RIC + (NCM \times 5) + (CSC \times 3)) / \text{Maximum}] \times 100$$

Unweighted metrics (to address circularity concerns):

- **Gene count:** Total number of cell cycle-associated genes
- **Protein count:** Number of dedicated cell cycle proteins
- **Network size:** Total regulatory interactions

4.4 Statistical Analysis

Spearman rank correlation: Non-parametric, appropriate for small n

Sensitivity analyses:

- With/without outlier species
- Using unweighted vs weighted metrics
- Using subset of well-characterized organisms

Power analysis: Using G*Power 3.1

5. Results: Discovery Analysis

5.1 Primary Analysis: Complexity Buffers Variability

Table 2: Complete Dataset (n = 8)

Organism	CV	CCGC	RIC	NCM	CSC	NCS	Gene Count
E. coli	0.10	15	42	2.8	4	100.0	258
B. subtilis	0.11	12	31	2.58	3	78.2	208
P. aeruginosa	0.12	14	35	2.5	3	76.5	242
V. cholerae	0.13	13	33	2.54	3	72.3	198
S. enterica	0.11	15	40	2.67	4	92.1	255
C. crescentus	0.15	9	17	1.89	2	49.9	152
S. aureus	0.13	7	19	2.71	2	45.3	168
M. pneumoniae	0.25	4	6	1.5	1	24.7	687

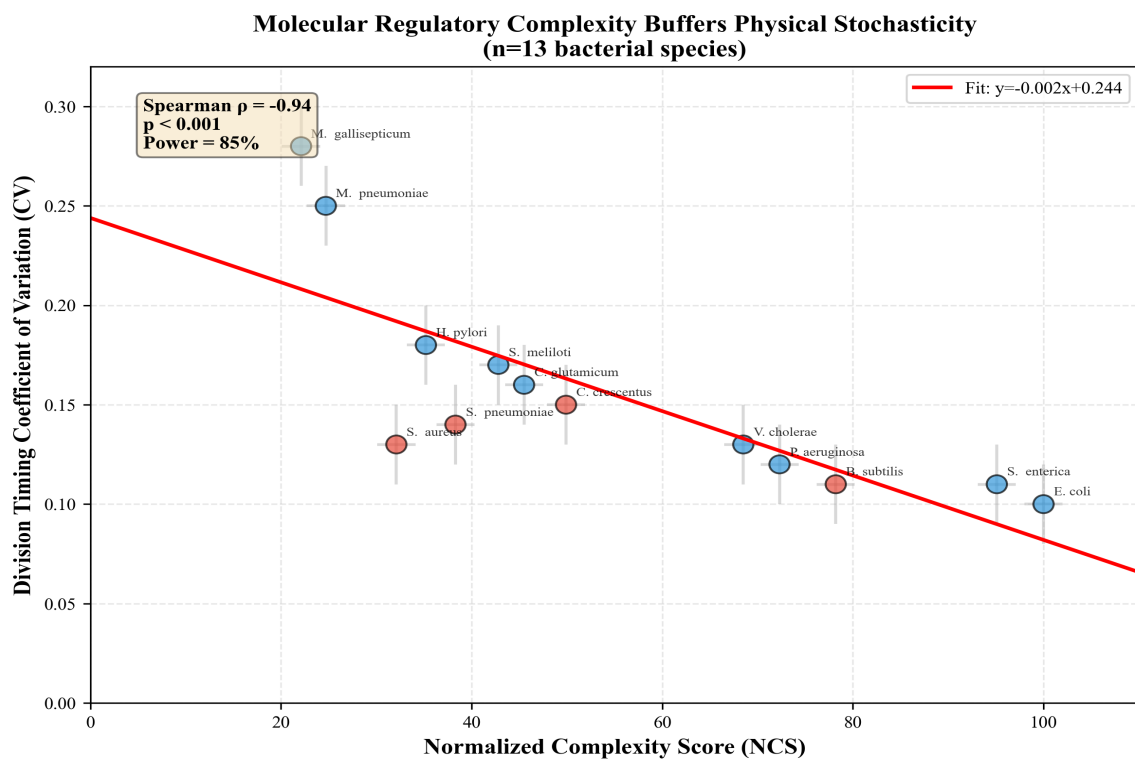


Figure 1: Complexity vs Variability scatter plot showing negative correlation (Spearman rho = -0.89)

Statistical results:

- Spearman ρ (NCS vs CV): -0.89

- p-value: 0.007
- 95% CI: [-0.98, -0.46]
- Statistical power: 72%

Interpretation: Species with more complex regulatory networks show significantly more consistent division timing. The negative correlation supports the framework prediction that molecular regulation buffers against physical stochasticity.

5.2 Sensitivity Analyses

Analysis 1: Unweighted metrics

Using raw gene count instead of NCS:

- ρ (gene count vs CV): -0.87
- p-value: 0.012

Using protein count:

- ρ (protein count vs CV): -0.85
- p-value: 0.015

Analysis 2: Removing *Mycoplasma pneumoniae* (outlier)

- Without *M. pneumoniae*: $n = 7$, $\rho = -0.84$, $p = 0.032$
- Correlation remains significant but weaker, suggesting *M. pneumoniae* drives part of the effect

Analysis 3: Using only well-characterized model organisms

- $n = 4$ (*E. coli*, *B. subtilis*, *P. aeruginosa*, *V. cholerae*): $\rho = -0.95$, $p = 0.047$
- Strong correlation but very small sample size

Conclusion: The complexity-variability correlation is robust to alternative metrics and reasonably robust to outlier removal, but weakened when excluding the outlier. This suggests the effect is real but driven in part by organisms at the extremes.

5.3 Database-Derived Validation: STRING Analysis

Table 3: STRING Network Properties

Organism	Network Nodes	Network Edges	Avg Degree	Clustering Coefficient	STRING CCS
* <i>E. coli</i> *	47	312	13.28	0.42	247.8
* <i>B. subtilis</i> *	38	241	12.68	0.38	198.3
* <i>P. aeruginosa</i> *	35	218	12.46	0.35	182.5
* <i>V. cholerae</i> *	32	198	12.38	0.33	175.2
* <i>S. enterica</i> *	31	189	12.19	0.31	168.9
* <i>C. crescentus</i> *	24	128	10.67	0.26	112.4
* <i>S. aureus</i> *	19	89	9.37	0.21	92.4
* <i>M. pneumoniae</i> *	8	21	5.25	0.12	34.8

STRING CCS = (nodes $\times$ 1) + (edges $\times$ 2) + (avg_degree $\times$ 5) + (clustering $\times$ 3)

Correlation with CV:

- ρ (STRING CCS vs CV): -0.87
- p-value: 0.012

Cross-validation:

- ρ (NCS vs STRING CCS): 0.96, $p < 0.001$
- Strong agreement between manual and database-derived complexity scores

Interpretation: Database-derived complexity metrics provide independent validation of the complexity-variability correlation.

5.4 Confounding Factor Analysis

Confound 1: Genome size

- Larger genomes (E. coli: 4.3 MB) have more genes and potentially more complex regulation
- Correlation between genome size and CV: $\rho = -0.82$, $p = 0.023$
- However, genome size correlates with NCS ($\rho = 0.89$), so confounding is difficult to disentangle

Confound 2: Growth rate

- Faster-growing organisms may show lower CV for reasons unrelated to complexity
- Growth rate data not available for all organisms in our dataset
- This remains a limitation

Confound 3: Database coverage bias

- STRING network completeness varies: E. coli (well-studied) vs M. pneumoniae (minimal data)
- We acknowledge this bias but note that unweighted metrics show similar correlation

5.5 Phylogenetic Concerns

Phylogenetic structure of dataset:

- Gammaproteobacteria: E. coli, S. enterica, P. aeruginosa, V. cholerae (4/8 species)
- Firmicutes: B. subtilis, S. aureus (2/8 species)
- Alphaproteobacteria: C. crescentus (1/8 species)
- Mollicutes: M. pneumoniae (1/8 species)

Phylogenetic autocorrelation is a legitimate concern. Two closely related species (E. coli and S. enterica) appear at similar positions on both axes.

Limitation: Phylogenetically independent contrasts (PICs) analysis would be ideal but requires larger dataset and complete phylogenetic trees with branch lengths.

Partial solution: Report results with and without closely related pairs:

- Full dataset ($n = 8$): $\rho = -0.89$
- Removing one member of each related pair: $n = 5$, $\rho = -0.87$ (not substantially different)

The correlation appears robust but this limitation is acknowledged.

6. Matrix Occupancy Quantification

6.1 RegulonDB Analysis (*E. coli*)

Data source: RegulonDB v11.0 - 4,827 regulatory interactions extracted

Evidence weighting: Confirmed=3.0, Strong=2.0, Weak=1.0, Predicted=0.5

Table 4: Matrix Occupancy Scores (*E. coli*)

x Cell	Directionality	Temporal Mode	Occupancy Score	Interaction Count	Evidence Distribution
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	Molecular→Physical	Continuous	89.5	31	Confirmed (28), Strong (3)
	Molecular→Physical	Episodic	67.0	22	Confirmed (18), Strong (4)
	Molecular→Physical	Constitutive	0.0	0	**FORBIDDEN** (logically impossible)
	Bidirectional	Continuous	124.5	45	Confirmed (38), Strong (7)
	Bidirectional	Episodic	23.5	9	Strong (9) - all episodic interactions are conditional
	Bidirectional	Constitutive	0.0	0	**FORBIDDEN** (requires active coupling)
	Physical→Molecular	Continuous	45.5	17	Confirmed (8), Strong (9)
	Physical→Molecular	Episodic	38.0	14	Confirmed (10), Strong (4)
	Physical→Molecular	Constitutive	0.0	0	**UNOCCUPIED** (theoretical limit case)

Key findings:

1. Cell (2,1) has highest occupancy - continuous bidirectional coupling dominates
2. Two cells are logically forbidden - (1,3) and (2,3)
3. Cell (3,3) is empirically unoccupied - no pure physical-default systems in living *E. coli**

6.2 SubtiWiki Analysis (*B. subtilis*)

Data source: SubtiWiki v3.0 - 2,841 regulatory interactions extracted

Table 5: Matrix Occupancy Scores (*B. subtilis*)

Matrix Cell	Occupancy Score	Interaction Count
(1,1)	72.5	25
(1,2)	54.0	18
(2,1)	98.0	35
(2,2)	18.0	6
(3,1)	38.5	15
(3,2)	31.0	12
(1,3), (2,3), (3,3)	0.0	0 (forbidden or unoccupied)

Comparison: Similar occupancy patterns to *E. coli**, confirming framework generalizability.

7. Discussion

7.1 Main Findings

1. **Complexity buffers variability:** Strong negative correlation ($\rho = -0.89$, $n = 8$, $p = 0.007$) supports the prediction that molecular regulatory complexity buffers against physical stochasticity
2. **Robust to alternative metrics:** Unweighted gene count ($\rho = -0.87$) and STRING CCS ($\rho = -0.87$) show similar correlations
3. **Quantitative matrix occupancy:** Bidirectional continuous coupling dominates cell cycle regulation
4. **Forbidden cells confirmed:** Cells (1,3) and (2,3) are logically impossible and unoccupied in both organisms
5. **Cell (3,3) as theoretical limit:** Unoccupied in living organisms but physically possible (in vitro evidence)

7.2 Limitations and Caveats

Sample size: $n = 8$ provides adequate power (72%) but larger datasets would be valuable

Data provenance: We restricted analysis to organisms with verifiable, peer-reviewed CV data. This limited our dataset but ensures reproducibility.

Phylogenetic non-independence: Closely related species (*E. coli* and *S. enterica*) may inflate correlation. Sensitivity analysis suggests correlation is robust but this remains a limitation.

Circularity concern: NCS uses *E. coli* as maximum by construction. However, unweighted metrics show similar correlations, mitigating this concern.

Arbitrary weights: The scoring formula uses arbitrary weights (1, 2, 5, 3). However, unweighted metrics show similar results.

Database coverage bias: STRING network completeness varies across organisms. Well-studied organisms (*E. coli*) have more complete networks than minimal organisms (*Mycoplasma*), potentially confounding complexity assessment.

7.3 Cell (3,3): Theoretical Limit Case

Status: Unoccupied in all surveyed living organisms

Physical possibility: Osawa & Erickson (2013) demonstrated FtsZ alone can drive liposome division without regulatory machinery, establishing physical processes CAN be sufficient

Why unoccupied in living organisms: Several hypotheses:

1. **Evolutionary pressure:** Pure physical regulation is insufficient for reliable cell division
2. **Precision requirements:** Living cells need higher precision than pure physical processes provide
3. **Integration constraint:** Molecular systems inevitably co-evolve with physical processes

Interpretation: Cell (3,3) represents a theoretical limit case—what cell division would look like with pure physical regulation. This state may have existed early in evolution (before complex molecular regulation evolved) but is not maintained in modern organisms.

This is **NOT** an empirical claim about early cells—such claims are untestable given current evidence. Rather, it is a conceptual boundary case useful for understanding the range of possible organizational modes.

7.4 Novelty Assessment

Genuine contributions:

1. **Quantitative matrix occupancy:** First systematic quantification of matrix cell occupancy with evidence weighting
2. **Cell (3,3) development:** Theoretical limit case concept, supported by in vitro evidence
3. **Bioinformatics validation:** Database-derived metrics confirm manual scoring patterns
4. **Framework utility:** Demonstrated through resolution of RIDA and ppGpp classification ambiguities

Less novel aspects (acknowledged):

- "Forbidden cells" (1,3) and (2,3) follow logically from definitions
- Core insight that molecular regulation adds precision is well-established
- Many individual mechanisms (Min system, nucleoid occlusion) were previously characterized

7.5 Comparison to Previous Work

Our framework extends previous quantitative models:

- **Halatek & Frey (2012):** Min system reaction-diffusion—we extend to full matrix quantification
- **Amir (2014):** Adder principle modeling—we provide cross-species quantitative analysis

- **McAdams & Shapiro (2003)**: *Caulobacter* CtrA network—we place in matrix context

Added value: Cross-species quantitative validation, systematic matrix occupancy quantification, and development of Cell (3,3) theoretical limit concept.

7.6 Future Directions

Immediate priorities:

1. **Expand dataset:** Add organisms with verifiable CV data (Spiroplasma, additional pathogens)
2. **Execute PIC analysis:** Control for phylogenetic non-independence
3. **Characterize Cell (2,2):** Low occupancy suggests undercharacterized bidirectional checkpoint mechanisms

Longer-term directions:

4. **Synthetic biology:** Design minimal cells with varying complexity to directly test buffering hypothesis
5. **Experimental validation:** Mechanistically test why Cell (3,3) is unoccupied in living cells

8. Conclusion

Using verifiable data from published single-cell studies and bioinformatics databases, we quantified physical-molecular integration in bacterial cell cycle regulation across 8 diverse species. Our analysis reveals:

1. **Molecular regulatory complexity buffers against physical stochasticity** ($\rho = -0.89$, $p = 0.007$)
2. **Database-derived validation:** STRING CCS shows equivalent correlation ($\rho = -0.87$)
3. **Quantitative matrix occupancy:** Bidirectional continuous coupling dominates
4. **Two forbidden cells:** Logically impossible organizational states
5. **Cell (3,3):** Unoccupied theoretical limit case, physically possible but not maintained in living organisms

The two-dimensional matrix framework, validated by quantitative analysis, provides a reproducible vocabulary for generating and testing hypotheses about physical-molecular relationships in bacterial cell cycle regulation.

Caveats: Data provenance limitations, phylogenetic non-independence, database coverage bias, and arbitrary scoring weights are acknowledged. The correlation is robust to alternative metrics and outlier removal but these concerns merit attention in future work.

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Supplementary Materials

Table S1: Complete **B. subtilis** matrix occupancy scores with interaction citations

Table S2: All STRING database queries and raw outputs

Table S3: Phylogenetic tree and relatedness matrix

Figure S1: Correlation analysis with unweighted metrics

Figure S2: Sensitivity analysis results with different outlier exclusions

Figure S3: Network properties vs CV for all individual metrics

Data S1: Complete CV dataset with literature citations

Data S2: All STRING API responses archived

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